



Zachary Treisman
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Academic Positions Held

- **Assistant Professor, Western Colorado University.** July 2019–

I am the primary data analysis and statistics person on campus. I oversee our Data Analytics minor, teach (or mentor the teachers of) our statistics courses, consult with faculty across disciplines and serve the university on committees which require sophisticated data analysis. In addition, I teach courses across the mathematics curriculum.

- **Honors Program Director, Western Colorado University.** January 2022–

Western Colorado University's Honors Program is an academic and social program that enrolls approximately 300 students. Faculty from across campus teach in the program, which consists of a combination of honors specific courses and honors versions of courses in any discipline. Students in the program have access to honors specific housing and special social and academic events. As the program director, I am responsible for all aspects of the program, including budgets, setting curriculum and recruiting faculty, recruiting and advising students, as well as teaching honors courses.

- **Visiting Professor/ Lecturer, Western Colorado University.** July 2009–June 2019

I came to Western Colorado University for a non tenure-track position. It turned out that I liked the location and my colleagues enough to stay indefinitely. During my time as a non tenure-track faculty member, I taught nearly all of the mathematics courses offered at this university, and my primary area of inquiry shifted from algebraic geometry to statistics and data analysis.

- **Postdoctoral Fellow, University of British Columbia.** January 2008–June 2009

This was a research and teaching appointment. I worked with Julia Gordon on number theoretic applications of p -adic and motivic integration, and taught large format courses on Multivariable Calculus and Linear Programming.

- **Visiting Part-Time Lecturer, University of Washington.** September–December 2007

This was a teaching appointment. I taught Differential Equations and Advanced Multivariable Calculus.

- **Visiting Fellow, Tata Institute of Fundamental Research.** September 2006–August 2007

This was a research appointment. I worked with Srinivas on algebraic geometry related to the minimal model program.

Consulting and non-academic employment

- **Data Analytics/ Machine Learning/ Statistical consultant, Gunnison Valley Cold Air Pooling study.** 2022–

Using machine learning and statistical models based on weather data acquired by a network of sensors placed at topographically strategic locations around the valley I am working with a team to understand the phenomenon of cold air pooling, where certain meteorological and topographic conditions create microclimates with significant temperature differentials from the surrounding area.

- **Data Analytics/ Machine Learning consultant, San Juan National Forest Beetle and Fire Recovery Study.** 2021–

Using machine learning models based on a combination of vegetation surveys gathered in field studies and publicly available satellite data, I am working with Jonathan Coop and Kyle Rodman to understand how various factors influence forest recovery after catastrophic fires and widespread beetle infestations.

- **Data Analytics/ Machine Learning consultant**, *Gunnison Valley Winter Backcountry Recreation Study*. 2020–

Winter backcountry recreation creates significant traffic on public lands in the Gunnison Valley, which includes the town of Crested Butte and surrounding areas. Several faculty and graduate students have been working with the US Forest Service to assess various aspects of this usage. I have been involved in the context of analyzing data collected from cameras located at various trailheads, correlating usage by various user types (cross-country skiers, snowmobilers, etc.) with variables such as weather conditions and avalanche danger ratings, as well as using machine learning to automate the classification of the images obtained by the trail cameras.

- **Data Analytics/ Statistical consultant**, *Western Colorado University Salary Assessment Task Force*. 2021–2022.

In this study we addressed the perception that employees of the university are underpaid, and more importantly that there exists pay inequity based on inappropriate variables such as gender. This required comparing the university’s data to an appropriately defined peer group for which data are available, and adjusting for appropriate variables such as cost of living. The data were compelling enough that salaries were and continue to be adjusted by providing appropriate raises with the goal of reducing and eliminating the discovered deficiencies.

- **Data Analytics/ Statistical consultant**, *Western Colorado University Strategic Resource Allocation*. 2019–2020.

In this study we analyzed all programs on campus for the purpose of ensuring that the university’s limited resources were being utilized consistent with its mission, in support of its strategic plan and to the maximum benefit of its students, faculty and staff. The study resulted in significant allocation adjustments in the form of both reductions and additional investments.

- **Statistical/ mathematical consultant**, *Ramah v. Interior*. 2013–2016.

I worked for the legal team representing a class of over 500 tribes and tribal organizations to assist in the settlement process after this case had been decided by the United States Supreme Court. The court’s decision in favor of the tribes required the government to pay damages appropriate to 19 years of accrued unpaid contract support costs on tens of thousands of contracts. Since a full audit was impossible, this was a sampling and estimation exercise. On this project I worked alongside statistician Michael Larsen, who was a professor at Georgetown at the time and also a consultant with the US Census Bureau. In addition to communications with the legal team representing the tribes, with the accountants who were also consulting for the tribe’s lawyers and with Dr. Larsen, this job involved extensive interactions with the Justice Department’s legal, accounting and data analytics teams, as well as communications with the clients directly. In short, I turned data into conversations with lawyers and policy-makers. The final settlement was approximately \$1 billion. See <http://www.scotusblog.com/case-files/cases/salazar-v-ramah-navajo-chapter/>.

- **Knowledge Engineer** at Kindred Communications. 1999.

This was my first job out of college. Kindred Communications was a Microsoft Vendor working with a custom Bayesian Network tool to build the logical engine for the help system in Windows Millenium. I was good at algebra, probability and graph theory, and found myself quite valuable to the work in the short time that I was there. When I started having questions that nobody at Kindred Communications could answer, I visited with David Heckerman at Microsoft Research and had some productive conversations. As I recall he suggested that I come work with him on some other projects. Unfortunately for my future career, my 22 year-old self had other plans, and I left this opportunity after only a few months to spend the rest of my gap year between college and graduate school driving a taxi in Santa Fe, New Mexico.

Computer Programs and Languages

- I mostly work in **R**. Frequently used packages include reticulate to access Python for advanced machine learning models; other R-native machine learning packages like RandomForest; ggplot2, dplyr and the rest of the tidyverse; glmmTMB for generalized linear mixed models; brms for Bayesian models; terra for spatial data; and TDA for topological data analysis.

- I've done several projects directly in **Python** and am familiar with pandas, numpy, matplotlib, tensorflow, networkx, and other frequently-used libraries.
- I'm comfortable with **git**.
- I can write scripts in **SQL**.
- I know my way around a command line and can communicate with **AWS** and **Azure**.
- For appropriate small-data and collaborative tasks, I'm quite capable with **Excel**.
- I'm fluent in **Mathematica**, **Maple**, **Sage** and **Matlab**.
- I've dabbled with **Julia**, **Rust** and **Haskell**.
- While I'm comfortable with all the usual software that everyone uses, I set my machine up to dual boot as a **Linux** machine and often prefer working in that environment.
- As a mathematician, I've used all sorts of specialized scripting/programming tools like **gap**, **cocoa**, **surf** and other languages that no longer exist or have been absorbed into larger systems.
- This document and many things I write are typeset in L^AT_EX.
- In the long ago days I wrote programs in **C**.

Education

- **Ph.D in Mathematics, University of Washington.** 2006.
- **M.S. in Mathematics, University of Washington.** 2003.
- **Santa Fe Institute Complex Systems Summer School.** 2001.
- **B.A. in Mathematics, Reed College.** 1999.
- **Budapest Semesters in Mathematics.** 1998.

Grants

- **NRT-HDR: Convergent Research in Integrative and Public Land Management**, *NSF 21-536 National Science Foundation Research Traineeship (NRT) Program*

Publications

- **The Boundary of the Visible.** *TSAIBERNETICS: transgenerational cybernetic art* — Wen-Ying Tsai, London Tsai, Ming Tsai. Laznia Centre for Contemporary Art, 2023
- **To the World's End/ A Circle Bundle Over a Circle.** *Bridges Stockholm 2018: Proceedings*, <http://www.bridgesmathart.org>
- **A Young Person's Guide to the Hopf Fibration.** <https://arxiv.org/abs/0908.1205>
- **Rationally generated arcs.** *Journal of Geometry*, Volume 90, Numbers 1–2 / December, 2008
- PhD Thesis: **Arc spaces and rational curves**
- Master's Thesis: **Some ideas regarding Kobayashi hyperbolicity and algebraic hyperbolicity of varieties over $\mathbb{C}$.**
- Undergraduate Thesis: **Tile patterns on the hyperbolic plane.**

Teaching

- Western Colorado University: 8-15 credits per semester. Classes include:

– Abstract Algebra – Advanced Ecological Data Analysis – Analysis – Calculus I – Calculus I (Honors) – Calculus III – College Algebra – History of Mathematics – Honors: The Shape of Space – Honors Forum	– Honors Gateway – Honors Oxford Tutorial – Intro. to Advanced Mathematics – Intro. to Complex Analysis – Peer Tutoring in Mathematics – Precalculus – Probability and Statistics – Quantitative Literacy – Secondary Mathematics – Mathematics Senior Seminar
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- **Special Topics: Dimension**
- **Statistical Modeling and Simulation**
- **Topics in Geometry**

- Summer Institute for Mathematics at the University of Washington (SIMUW): Two week mini-courses on **The Shape of Space**, **The Hopf Fibration** and **Non-Euclidean Geometry**.
- UBC: one class per semester. Classes taught: **Calculus III** and **Linear Programming**.
- UW: One class (7 quarters) and two classes (2 quarters). Classes taught:

- **Advanced Multivariable Calculus**
- **Business Algebra**
- **Business Calculus**
- **Calculus I**
- **Geometry for High School Teachers**
- **Introduction to Differential Equations**
- **Linear Algebra**
- **Linear Analysis**

University Service

- GE Committee, 2024–present.
- Headwaters/Delta Task Force, 2024–present.
- Core Liberal Arts & Sciences Task Force, 2023–2024.
- Curriculum Committee, 2020–present.
- Honors Council, 2019–2022.
- Mathematics work-study coordinator, 2011–2018.
- Senior seminar mentoring. Typically one student per semester, 2009–present.
- IT Committee. 2012–2018. (Chair 2013–2015)
- Assessment Committee. 2011–2013

Invited Talks

- **The vague boundary between geometric models and mathematical art: Examples from the geometry classroom to the sculpture gallery.** Joint Mathematics Meetings, Denver 2020.
- **To the World's End/ A Circle Bundle Over a Circle.** Bridges Stockholm 2018.
- **Mapmaking and vice-versa.** Adams State University, 2016; Reed College, 2014; Shift Studio Gallery *Dialogues with Artists and Mathematicians*, Seattle, WA, 2007.
- **Image and pattern in hyperbolic geometry.** Plenary lecture at Mathday 2010, University of Washington; Adams State College, 2010; Colorado College, 2009; Western State College, 2009.
- **Images of hyperbolic geometry.** SIMUW, 2005, 06; Mathday 2004, 06, 08, 09.
- **Rationally generated arcs.** University of British Columbia, 2007; Institute of Mathematical Sciences, Chennai, India, 2006.
- **Arc spaces in algebraic geometry.** Chennai Mathematical Institute, India, 2006.
- **Illustrating mathematics with computers.** Institute of Mathematical Sciences, Chennai, India, 2006.
- **Arc spaces and rational curves.** Freie Universität, Berlin, Germany, 2006.
- **Algebraic geometry for the jet set.** Reed College, 2004.

Art and Mathematics

- *Rule of Law.* (A section of a Hopf fibration) With London Tsai.
- *To the World's End.* (A circle bundle over a circle) With London Tsai. <http://www.londontsai.com/qr/worldsend.htm>
- *The Hopf map, a flipbook.* (An animation of a Hopf fibration) With Matt Kahle.
- *Hyperbolic Immersion.* (A model of non Euclidean geometry)
- *Digital Sundial.* With Jordan Schwartz.

Other notes and past projects

- Board of Directors, Mountain Roots Food Project. 2011–2016.

- Volunteer, Gunnison Trails. 2013–present.
 - TOPS Elementary/Middle School. Math Olympiad Club mentor, 2004–2006.
 - Solo summit ski of Denali (Mt. McKinley) via West Buttress and West Rib.
 - WFR (Wilderness First Responder) and AIARE level 2 (Avalanche hazard analysis) certification.
 - EMT-B certification
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